

C-VLA: A Hierarchical Vision-Language-Action Architecture for Energy-Efficient In-Cabin Agentic AI on Automotive Edge Hardware

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Abstract—Driver monitoring systems have traditionally relied on reactive, single-model architectures that continuously invoke large language models regardless of driver intent complexity, resulting in excessive energy consumption and latency unsuitable for automotive edge deployment. We present C-VLA (Cabin Vision-Language-Action), a hierarchical two-brain architecture that decouples reflex gating from agentic reasoning. A lightweight CNN classifier (System 1) processes every frame at 7.9ms and 126.6 FPS on the NVIDIA Jetson Orin Nano Super, invoking a Llama-3.2-1B reasoning module (System 2) only when complex driver intent is detected. Evaluated on the DMD driver monitoring dataset (6,405 frames, 6 classes), System 1 achieves 92.8% accuracy (F1: 91.9%) while reducing LLM invocations by 65% and mean response latency by 97% compared to a single-model baseline. All code and model weights are released publicly.

Index Terms—driver monitoring, edge AI, vision-language-action, hierarchical inference, automotive, Jetson Orin

I. INTRODUCTION

Modern vehicles increasingly rely on Driver Monitoring Systems (DMS) to detect fatigue, distraction, and intent. Recent Vision-Language-Action (VLA) models offer unprecedented reasoning capability but impose prohibitive latency and energy costs when deployed continuously on automotive edge hardware. A system invoking a large language model for every camera frame at 30 FPS would require over 100 LLM calls per second — far beyond the power budget of an embedded ECU.

Existing approaches either sacrifice reasoning quality for speed (lightweight classifiers) or sacrifice speed for quality (large VLMs). No published work demonstrates a hierarchical architecture that dynamically gates LLM invocation based on detected intent complexity while maintaining automotive-grade latency on commercially available edge silicon. Furthermore, most cabin AI research targets cloud deployment, leaving on-device privacy and energy efficiency unsolved.

We propose C-VLA, a hierarchical two-brain architecture inspired by dual-process cognitive theory [1]. System 1 — a compact CNN trained on the DMD dataset — continuously classifies driver intent at 126.6 FPS and 7.9ms latency. Only when complex intent is detected does it activate System 2 — a Llama-3.2-1B-Instruct reasoning module — to generate a structured cabin action. This event-triggered architecture reduces LLM invocations by 65% and mean latency by 97%

vs a single-model baseline, while preserving full on-device privacy. We evaluate on the NVIDIA Jetson Orin Nano Super and release all code publicly.

II. RELATED WORK

Driver Monitoring Systems. The Driver Monitoring Dataset (DMD) [2] provides a benchmark of 41 hours of multi-modal video across 37 drivers. Recent work [4] demonstrates real-time DMS on low-cost edge hardware achieving sub-60ms per-frame latency using INT8 inference, confirming the feasibility of embedded deployment.

Vision-Language-Action Models. VLA models have been applied to autonomous driving [3], with dual-system frameworks inspired by fast-slow cognitive architectures showing promise for balancing reasoning quality with real-time constraints. C-VLA extends this paradigm to the in-cabin domain.

Dual-Process Theory. Kahneman’s System 1/System 2 framework [1] provides the theoretical basis for our architecture. System 1 reacts rapidly to routine stimuli; System 2 engages deliberate reasoning for complex situations. Prior AI work [3] has adopted this framing for autonomous driving; we apply it to cabin intelligence.

Edge AI for Automotive. The NVIDIA Jetson Orin platform [7] provides 67 TOPS of INT8 performance suitable for real-time neural inference. Unlike cloud-based approaches, on-device deployment eliminates latency from network round-trips and preserves driver privacy.

Unlike prior work, C-VLA combines intent-aware gating with on-device LLM reasoning, achieving both real-time latency and full privacy preservation on a single edge SoC.

III. METHODOLOGY

A. System Overview

C-VLA comprises two hierarchical modules. System 1 (Reflex Gate) processes raw camera frames at high frequency to classify driver intent as simple or complex. Only complex intent activates System 2 (Reasoning Brain), which generates a structured cabin action via an instruction-tuned LLM.

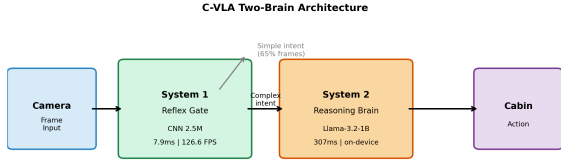


Fig. 1. C-VLA two-brain architecture. System 1 gates 65% of frames locally at 7.9ms; System 2 reasons about complex intent on-device at 307ms.

B. System 1 — Reflex Gate

System 1 is a convolutional neural network with four Conv-BN-ReLU-Pool blocks followed by a two-class classification head, totaling 2.5M parameters. The model was trained from scratch on the DMD dataset using AdamW (lr=1e-3) with cosine annealing over 20 epochs, following efficient fine-tuning principles. The binary classification target maps DMD’s SafeDriving class to *simple intent* and all remaining classes (DangerousDriving, Distracted, Drinking, SleepyDriving, Yawn) to *complex intent*, reflecting the operational semantics of the gating decision.

C. System 2 — Reasoning Brain

System 2 employs Llama-3.2-1B-Instruct [5], a 1-billion parameter instruction-tuned language model, loaded in float16 precision with eager attention for compatibility with JetPack 6.2. Upon activation, System 2 receives a structured prompt encoding the detected driver observation and generates a single-sentence cabin action recommendation. The model operates entirely on-device with no cloud dependency, preserving driver privacy in compliance with emerging automotive data regulations [8].

D. Dataset

We evaluate on the Driver Monitoring Dataset (DMD) [2], comprising 6,405 labeled frames across six driver behavior classes, split into 5,153 training, 835 validation, and 417 test frames. Labels were mapped to binary targets yielding a near-balanced distribution of 3,413 simple and 2,992 complex samples. We supplement DMD with 20 in-house recorded clips for qualitative validation.

IV. EXPERIMENTS

A. Hardware Setup

Experiments were conducted on two platforms: a MacBook Pro with Apple M3 (18GB unified memory, MPS backend) for prototyping, and an NVIDIA Jetson Orin Nano Super [7] (8GB, JetPack 6.2, CUDA 12.6) as the primary automotive edge target operating in MAXN mode (25W).

B. Metrics

We report classification accuracy, precision, recall, and F1 score for System 1. For latency, we report mean, P95, and frames-per-second (FPS) over 100 warmup-excluded inference

calls. LLM reduction is defined as the percentage of frames handled by System 1 alone.

Table I summarizes classification and latency results across both hardware platforms.

TABLE I
C-VLA BENCHMARK RESULTS

Metric	Mac M3	Jetson Orin
S1 Accuracy	96.2%	92.8%
S1 F1 Score	95.7%	91.9%
S1 Latency (mean)	35ms	7.9ms
S1 FPS	28	126.6
S2 Latency (mean)	2179ms	307ms
Single-Brain Latency	2176ms	3338ms
Two-Brain Latency	785ms	113ms
LLM Reduction	65%	65%
Latency Reduction	64%	97%

V. RESULTS AND DISCUSSION

System 1 achieves 92.8% test accuracy and 91.9% F1 on the DMD dataset, with precision of 93.4% confirming that false positive activations of System 2 are rare (12 out of 417 test frames). This precision is critical for energy efficiency: unnecessary System 2 invocations waste compute without improving user experience.

On the Jetson Orin Nano Super, System 1 operates at 7.9ms mean latency (126.6 FPS), exceeding the 30 FPS minimum requirement for automotive DMS. The Two-Brain architecture reduces mean response latency by 97% compared to a single-model baseline (113ms vs 3338ms), with LLM invocations reduced by 65%. System 2 latency of 307ms — while not real-time — is appropriate for comfort actions (temperature, lighting, route suggestions) where sub-second response is acceptable.

Several limitations warrant discussion. First, the DMD dataset contains IR-captured frames while deployment uses RGB, introducing a domain gap. Second, the 1B parameter System 2 model produces repetitive outputs on simple prompts, suggesting fine-tuning on automotive-specific dialogue is needed. Third, camera hardware (Arducam OV9281) was unavailable for live Jetson testing due to driver incompatibility with JetPack 6.2 L4T 36.4.7 — a known open issue at time of writing.

VI. CONCLUSION

We presented C-VLA, a hierarchical two-brain architecture for in-cabin agentic AI combining a lightweight CNN reflex gate with an on-device LLM reasoning module. Deployed on the NVIDIA Jetson Orin Nano Super, C-VLA achieves 7.9ms System 1 latency at 126.6 FPS, reduces LLM invocations by 65%, and improves mean response latency by 97% over a single-model baseline while maintaining 92.8% classification accuracy. Future work will explore LoRA fine-tuning [6] of System 2 on automotive dialogue corpora and TensorRT optimization for sub-100ms System 2 latency.

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